MATHEMATICAL OPTIMIZATION: QUESTION LIST

Abstract. Some notes on organization and a list of questions helping you to prepare for the exam. At the exam, write formulas, write definitions, explain concepts.

Organization

- 2h lectures + 2h Lab exercises per week.
- 80% lab attendance to pass the exercises.
- Exam: 2h written exam; at most 1 retake. Copying from your neighbor result in a fail.
- Bring your laptop to the lab.
- Office hours: Monday 2-3pm, Room 4.19
- felix.huber@ug.edu.pl to contact me by email.
  Write class (MathOpt) and surname in subject.

Material

(1) Lovász, lecture notes: Semidefinite programs and combinatorial optimization `https://sites.math.washington.edu/~thomas/teaching/m514_web/Lovasz_semidef.pdf`
(2) Gupta, O'Donnel - CMU's course on LP and SDP (our convention) `https://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15859-f11/www/notes/lpsdp.pdf`
(3) Matoušek, Gärtner - Understanding and using Linear Programming `https://link.springer.com/content/pdf/10.1007/978-3-540-30717-4.pdf`
(4) Heng Yang - Semidefinite Optimization and Relaxation `https://hankyang.seas.harvard.edu/Semidefinite/` (careful: different convention for primal and dual)
(5) Blekherman et al, Semidefinite Optimization and convex algebraic geometry `https://www.mit.edu/~parrilo/sdocag/`
(6) JuMP library for Optimization `https://jump.dev/`

For all questions use notation/definitions from our class.

1. Linear Algebra

1. Give definitions for the scalar product, outer product for vectors. Trace inner product for matrices.
2. Define trace/rank of a square matrix.
3. What is a symmetric matrix, orthogonal matrix?
4. What is an eigenvector, eigenvalue; what are their properties for symmetric or positive semidefinite matrices?
5. What is the spectral decomposition? What is the Cholesky factorization?
6. Give five equivalent definitions for a matrix to be positive semidefinite.
7. For a matrix, what is the difference between positive and positive semidefinite?
8. What is the Löwner order?
9. What is a Gram matrix?
10. What is the positive cone?
11. Define convex cone, convex hull, dual cone, self-dual cone. Examples of self-dual cones?

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12. Show that $\sum_i p_i v v^T \succeq 0$ for all $v \in \mathbb{R}^n$ and $p_i \geq 0$.

2. Linear Programming

13. Formulate the standard form of a linear program and its dual. What are its parameters, variables, objective functions? What "sizes" do they have?
14. What is a feasible point, a feasible set?
15. What is a polytope? Give examples.
16. What is the non-negative orthant? The probability simplex? The unit simplex?
17. Explain Farkas Lemma and its use.
18. For an LP, what is weak duality, strong duality, duality gap?
19. Prove weak duality.
20. What is the duality theorem.
21. What is complementary slackness?
22. Assume the primal is feasible (infeasible, unbounded, optimal). What can we say about the dual (feasible/infeasible/optimal/unbounded)?
23. Explain s-t flow (max perfect matching, vertex cover); formulate an LP relaxation. What can we say about the relaxation?
24. What is an stable (or independent) set? What is the stable set polytope? What linear inequalities are known to bound it?
25. What is STAB, FSTAB, QSTAB? Draw the sets for the triangle and the line graph of three vertices.
26. What is a Chvátal-Gomory Cut? What is the Chvátal rank?
27. Write the following LP in primal and dual form:

$$\begin{aligned} \min \quad & x + y + z \\ \text{s.t.} \quad & x - y = 0 \\ & z + x \leq 1 \end{aligned} \tag{1}$$

What can you say about its primal and dual solution?

28. I have an LP but my solver tells me it is not feasible. How do I get an infeasibility certificate?

3. Semidefinite Programming

29. Formulate the primal and dual standard formulations of semidefinite programs. What are the problem parameters?
30. For an SDP, what is weak duality, strong duality, duality gap? Prove weak duality.
31. Complementary slackness?
32. What is a feasibility problem? What is an infeasibility certificate?
33. What is a spectrahedron? What is an elliptope?
34. How do I convert an matrix equality into an matrix inequality constraint, and vice versa?
35. Write in standard form, formulate dual, and solve by hand

$$\begin{aligned} \max \quad & 2x_{11} + 2x_{12} \\ \text{s.t} \quad & x_{11} + x_{22} = 1 \\ & \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \succeq 0 \end{aligned} \tag{2}$$

36.

$$\begin{aligned} \text{find} \quad & X \succeq 0 \\ \text{s.t} \quad & \langle A_1, X \rangle = 1 \\ & \langle A_2, X \rangle = -1 \\ & \langle A_3, X \rangle = 1 \end{aligned} \tag{3}$$

where

$$A_1 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

4. Binary quadratic optimization

37. What is the Ising model? What is Max Cut? How are they related?
38. Derive SDP the relaxation for Max Cut.
39. Formulate the binary quadratic optimization problem, and write the equivalent polynomial formulation.
40. In general, how can one get lower and upper bounds on a binary quadratic optimization problem?
41. Formulate SDP bounds on binary quadratic optimization problems (both primal and dual formulations). Explain the primal, derive its dual.
42. What is an approximation ratio?
43. Let u, v be two vectors. Denote by R a random unit vector. Let $x_u = \text{sign}(r^T u)$ and $y = \text{sign}(r^T v)$ What is the probability that $x = y$?
44. How does the Goemans-Williamson algorithm work? Formulate the steps.
45. What is a diagonally dominant matrix? How do they relate to positive semidefinite matrices?
46. How can one expand a diagonally dominant form?
47. What approximation ratios does one obtain for $\max x^T Ax$ in the case of diagonally dominant / semidefinite matrices A?
48. What is the Grothendiek problem with rank constraint?

5. Lovasz ϑ number

49. What is an independent set?
50. What is the Hamming distance?
51. What is the repetition code, what are its parameters?
52. How is the Lovasz theta number defined as an SDP?
53. What is the strong graph product? What is $\vartheta(G \boxtimes H)$?
54. Compute the strong graph product for two small given graphs.
55. What is the Shannon capacity of a graph? How does the Lovasz ϑ bound it?
56. What is the capacity of the Pentagon? Give an example where it is achieved.
57. Define independence number, chromatic number. What are their computational complexities?
58. What is Knuth's sandwich theorem? Why is it interesting/surprising?

6. Positive polynomials

59. What is a polynomial ring? What properties does a ring satisfy?
60. What is a polynomial optimization problem?
61. Prove that every SOS polynomial can be written as $p(x) = m^T Am$ with m a monomial vector.
62. What is a SOS proof?
63. I want to prove that the polynomial $p(x, y, z) = x^2 + y^4 + z^2 + 2xy^2 - 2y^2 z - 2xz$ is non-negative on $\mathbb{R}$. I have access to a SDP solver, how can I show non-negativity?
64. What is a variety? Give a simple example and draw it.
65. Formulate the Schmüdgen positivstellensatz.
66. Formulate the Putinar positivstellensatz.
67. Formulate the moment relaxation for polynomial optimization.
68. Formulate the sum-of-squares relaxation for polynomial optimization.
69. Give an example of a non-negative polynomial that is *not* sum of squares. How can I prove its non-negativity?
70. Give the geometric-arithmetic mean inequality. Under what conditions does it hold?

71. What is the Newton polytope? How does one use it?
72. How is the Minowski addition of two sets defined?

7. Groups and representations

73. What is a group.
74. What is a representation.
75. List all elements of S_3, give two representations.